

ARTIFICIAL INTELLIGENCE – ASSESSMENT TOOL 3

U5-A3 REVERSE ENGINEERING

PROJECT TITLE: MEDISCAN: AI-BASED MEDICAL IMAGE ANALYSIS AND DISEASE DETECTION SYSTEM

Name: M Pradeepthi

Reg No: 192472387

Course Code: CSA1704

Project Overview

MediScan is an AI-based medical image analysis and disease detection system designed to assist healthcare professionals in examining medical images and identifying possible disease patterns. The system can analyze images such as X-rays, CT scans, MRI scans, or other supported medical images using image-processing and machine-learning techniques.

The system collects medical images, preprocesses them, extracts relevant visual features, and applies an AI model to classify or detect abnormal conditions. Based on the predicted result, MediScan can provide a disease label, confidence score, highlighted region of interest, and supporting information for medical review.

For this reverse-engineering assessment, MediScan is treated as the existing intelligent system to be analyzed. The reverse-engineered design identifies its major modules, data flow, machine-learning pipeline, decision logic, intelligent-agent components, and possible improvements. The assignment requires system decomposition, architecture reconstruction, algorithm identification, data-flow mapping, and redesign strategies.

1. System Decomposition & Analysis

MediScan can be decomposed into a set of interconnected modules that transform raw medical images into disease-related predictions and decision-support information. The objective is not to replace medical professionals but to support faster and more consistent analysis.

Major Components

- Medical Image Collection Module – collects medical images from imaging devices, hospital databases, PACS systems, or uploaded datasets.
- Image Preprocessing Module – performs resizing, normalization, noise reduction, contrast enhancement, format conversion, and quality validation.
- Feature Extraction Module – identifies useful image characteristics such as edges, textures, shapes, intensity patterns, and deep visual representations.
- Disease Detection Module – uses an AI or deep-learning model to classify images or detect abnormal regions associated with possible diseases.

- Result Classification Module – converts model outputs into disease categories, normal/abnormal results, or severity levels where supported.
- Intelligent Decision-Support Agent – analyzes prediction confidence and image context to recommend whether the case should be reviewed, prioritized, or monitored.
- Feedback Module – records verified outcomes and specialist feedback for evaluation and future model improvement.
- Dashboard and Reporting Module – presents uploaded images, prediction results, confidence scores, highlighted regions, and reports to authorized users.

Underlying AI Logic

- Historical medical images are associated with labels representing normal conditions, disease classes, or annotated abnormal regions.
- The preprocessing pipeline converts raw medical images into model-ready image tensors or feature representations.
- A deep-learning model learns visual patterns associated with different medical conditions.
- The model produces class probabilities, disease predictions, or detection results.
- A confidence threshold can be applied to identify uncertain cases requiring specialist review.
- The decision-support component combines model confidence, image quality, and predefined clinical workflow rules.
- Verified outcomes and expert feedback can later be used to measure accuracy, precision, recall, F1-score, sensitivity, specificity, and operational usefulness.

Key Constraints

- Medical images may contain noise, low contrast, artifacts, inconsistent resolution, or incomplete information.
- Incorrect predictions can create false alarms or fail to identify important abnormalities.
- The system should provide understandable outputs such as confidence scores and visual explanations where possible.
- Patient images and related information must be protected through authentication, authorization, encryption, and secure data handling.
- The architecture should support increasing image volumes and repeated prediction requests.

2. Architecture Reconstruction

The reconstructed architecture of MediScan follows a layered design in which medical images move through collection, preprocessing, feature extraction, AI-based disease detection, decision support, and result presentation. The modules communicate through structured data interfaces or APIs.

Reconstructed Architecture

Data flow: Medical Images → Image Ingestion → Preprocessing & Feature Extraction → AI Disease Detection Model → Result Classification → Decision-Support Agent → Dashboard/Report → Feedback → Model Improvement.

Module Interaction

- The image-ingestion module receives medical images from supported hospital systems, storage services, or authorized uploads.
- The preprocessing module validates and transforms incoming images before they reach the AI model.
- The feature-extraction stage produces useful visual representations for disease classification or abnormality detection.
- The AI module receives the processed image and returns predicted classes, probabilities, or detected regions.
- The result engine maps model outputs to understandable categories and confidence levels.
- The decision-support agent uses prediction information to prioritize review or generate an appropriate system response.
- The dashboard displays the image, predicted result, confidence, and relevant visual explanation to authorized users.
- The feedback module records expert validation or confirmed outcomes for future evaluation and improvement.

Intelligent Agent Interaction

- Monitoring Agent: watches incoming medical images and checks image availability and processing status.
- Image Analysis Agent: performs or requests preprocessing and AI-based image analysis.
- Prediction Agent: generates disease probabilities, classification results, or abnormality detections.
- Decision-Support Agent: evaluates prediction confidence and workflow rules to prioritize cases or request further review.
- Feedback Agent: records verified outcomes and specialist feedback for model evaluation.
- Coordination Agent: combines information from other agents and manages the overall processing sequence.

3. Algorithm Identification

Reverse engineering MediScan involves identifying the likely image-processing and machine-learning stages used to transform medical images into disease-related predictions. The exact algorithm cannot be established from the project title alone; therefore, the following is a proposed algorithmic design for MediScan.

Image Processing Pipeline

- Image Validation: check file format, image quality, size, and availability of required information.

- Noise Reduction: reduce unwanted noise or artifacts using suitable filtering techniques.
- Normalization: scale pixel values into a consistent range required by the model.
- Image Resizing: convert images to a fixed resolution for efficient model processing.
- Contrast Enhancement: improve the visibility of relevant structures when required.
- Data Augmentation: generate suitable training variations to improve model generalization.
- Train-Test Split: separate labeled image records into training, validation, and testing data.
- Model Training: train an image-classification or detection model using labeled medical images.
- Prediction: generate disease probabilities or abnormality detections for new images.
- Evaluation: measure performance using accuracy, precision, recall, F1-score, sensitivity, specificity, ROC-AUC, and confusion matrix.

Proposed Machine-Learning Algorithm

A suitable primary algorithm for MediScan is a Convolutional Neural Network (CNN) because medical images contain spatial patterns, textures, edges, and structures that can be learned directly from image data. CNNs use convolutional layers to extract increasingly complex visual features and can support classification tasks such as normal versus abnormal or multi-class disease detection.

- Input: preprocessed medical image X .
- Output: predicted disease class and confidence probability $P(\text{class} | X)$.
- The convolutional layers learn visual features from local regions of the image.
- Pooling or down-sampling layers reduce feature-map size while preserving important information.
- Higher layers combine learned features for final disease classification or detection.
- The final output probability is compared with a confidence threshold or workflow rule.
- For example, a high-confidence abnormal result can be marked for priority review, while a low-confidence result can be flagged for additional verification.

Decision-Making Logic

- If the predicted abnormality confidence is high, the system prioritizes the case for authorized medical review.
- If confidence is moderate, the system can recommend additional verification or comparison with clinical information.
- If confidence is low, the system can continue normal workflow while retaining the result for monitoring.
- The intelligent agent can combine model confidence with image quality, approved clinical context, and previous verification outcomes.

4. Data Flow & Process Mapping

The MediScan data flow begins with a medical image and ends with an AI-assisted disease analysis result. The processing pipeline can operate in batch mode for stored image collections or near-real-time mode when new images become available.

Data Flow

Medical Image → Data Collection → Distributed Storage → Image Processing → Feature Engineering → AI Prediction → Agent Decision → Report & Feedback

PySpark RDD Operations

- `map()` transforms each image record or metadata entry into a structured representation containing required attributes.
- `filter()` removes invalid, corrupted, unsupported, or low-quality records according to analysis criteria.
- `flatMap()` can expand image-related records when multiple regions, slices, or annotations are associated with one study.
- `reduceByKey()` can aggregate study-level statistics or repeated processing information when required.
- `join()` combines image metadata with labels, annotations, or approved related records.
- `groupByKey()` can organize image records by study or patient identifier where grouping is required, although efficient aggregation functions are preferred when possible.
- `sortByKey()` can rank cases according to prediction confidence, priority, or processing status.

Communication Between Components

- Image ingestion communicates validated image records to the processing layer.
- The distributed processing layer produces processed images or feature records for the prediction service.
- The prediction service returns disease probabilities and relevant model outputs to the result engine.
- The result engine communicates confidence and priority information to the decision-support agent.
- The decision-support agent communicates recommended actions or case priority to the dashboard and reporting layer.
- Feedback from expert review is returned to the analytics layer for evaluation and future model improvement.

5. Improvement & Redesign

The reverse-engineered MediScan system can be improved by introducing stronger distributed image processing, real-time monitoring, explainable AI, agent coordination, model monitoring, and optimized resource usage. These improvements can make the system more scalable, efficient, secure, and practical for healthcare deployment.

Proposed Improvements

- **Distributed Image Processing:** use scalable processing for large medical-image collections so records can be partitioned and analyzed efficiently.
- **Real-Time Image Monitoring:** process newly available images and trigger analysis without unnecessary delay.

- **Feature Store:** maintain reusable extracted features and metadata so repeated prediction requests do not require complete recomputation.
- **Explainable AI:** provide visual explanations such as highlighted regions or feature-importance information to support result interpretation.
- **Agent-Based Analysis:** allow specialized agents to monitor images, analyze data, generate predictions, prioritize cases, and evaluate outcomes.
- **Model Monitoring:** track prediction quality, confidence distribution, data drift, and changes in image characteristics.
- **Threshold Optimization:** select confidence thresholds that balance false positives, false negatives, review workload, and safety requirements.
- **Scalable Storage:** separate long-term image archives from frequently accessed operational data and organize large datasets efficiently.
- **Fault Tolerance:** use checkpointing, retries, and distributed processing mechanisms to reduce the impact of task or node failures.
- **Security and Privacy:** apply role-based access, encryption, secure APIs, audit logging, and controlled access to medical data.
- **Continuous Learning:** periodically retrain or evaluate the model using newly verified outcomes and expert feedback.

Redesigned Architecture

Sources Streaming / Batch → Distributed Data → Image Processing Pipeline → Feature Store → AI Detection + Explainability → Multi-Agent Layer → Report + Feedback

The redesigned architecture creates a feedback loop: medical image → distributed image analytics → disease prediction → explanation → intelligent decision support → expert-verified outcome → model and strategy improvement. This allows MediScan to move from a simple image-classification system toward a more proactive AI-assisted medical image analysis platform.

Expected Benefits

- Higher scalability for large medical-image datasets.
- Faster processing through distributed computation and efficient data pipelines.
- Earlier identification of images that may contain abnormal patterns.
- More explainable and actionable AI-assisted predictions.
- Better coordination between image analysis and case-review activities.
- Continuous improvement using verified outcomes and feedback.
- Improved operational efficiency by prioritizing relevant or high-confidence abnormal cases.
- Greater reliability through fault-tolerant distributed processing.

6. Reverse-Engineered Processing Workflow

Step 1: Image Acquisition – An authorized user, imaging device, hospital database, or connected source provides a supported medical image.

Step 2: Image Validation – The system checks file type, image size, completeness, quality, and processing eligibility.

Step 3: Preprocessing – Noise reduction, normalization, resizing, and enhancement are applied according to the selected analysis pipeline.

Step 4: Feature Learning – The CNN automatically learns relevant visual patterns from the processed image.

Step 5: Disease Prediction – The model produces predicted classes, probabilities, or abnormality indicators.

Step 6: Result Interpretation – The result engine converts raw outputs into understandable labels and confidence values.

Step 7: Intelligent Decision Support – The agent evaluates confidence and predefined workflow rules to determine case priority or need for additional review.

Step 8: Dashboard and Report – Authorized users view the image, result, confidence, explanation, and processing status.

Step 9: Feedback – Expert verification or confirmed outcomes are recorded for performance evaluation and future improvement.

Example Decision Rules

- IF image quality is below the required threshold, THEN request image verification or reprocessing.
- IF abnormality probability is above the high-confidence threshold, THEN mark the case for priority review.
- IF abnormality probability is within an uncertain range, THEN recommend additional verification.
- IF the image is classified as normal with acceptable confidence, THEN complete standard reporting workflow.
- IF expert feedback contradicts the AI output, THEN store the verified outcome for evaluation and future model improvement.

7. Performance Evaluation and Monitoring

A reverse-engineered medical image analysis system should be evaluated continuously because performance can vary with image source, disease prevalence, image quality, and changes in real-world data.

Important Evaluation Measures

- Accuracy: overall proportion of correct predictions.
- Precision: proportion of predicted positive cases that are correct.
- Recall / Sensitivity: ability to identify relevant positive cases.
- Specificity: ability to correctly identify negative or normal cases.

- F1-Score: balance between precision and recall.
- ROC-AUC: ability of the model to distinguish classes across different thresholds.
- Confusion Matrix: detailed view of true positives, true negatives, false positives, and false negatives.

Monitoring Requirements

- Monitor changes in input image quality and distribution.
- Track confidence-score patterns and uncertain cases.
- Compare AI predictions with verified outcomes.
- Detect data drift and possible performance degradation.
- Maintain audit logs for processing and authorized access.
- Review false-positive and false-negative patterns to improve thresholds and training data.

Conclusion

Reverse engineering MediScan shows how an AI-based medical image analysis and disease detection system can be decomposed into image collection, preprocessing, feature extraction, deep-learning prediction, result classification, intelligent decision support, and feedback modules. The reconstructed architecture demonstrates a clear flow from medical-image input to AI-assisted disease analysis and reporting.

Image-processing operations and distributed data pipelines can support large-scale analysis, while a Convolutional Neural Network can learn relevant visual patterns for classification or detection. Intelligent agents can monitor incoming images, coordinate analysis, prioritize results, and collect verified feedback.

The proposed redesign further improves scalability, explainability, reliability, security, and continuous learning. By combining AI prediction with controlled decision support and expert feedback, MediScan can serve as a practical foundation for an AI-assisted medical image analysis platform.